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| Exp6 (1).ipynb | Add files via upload | now |
| LICENSE | Initial commit | last year |
| README.md | Update README.md | 2 minutes ago |
| sobel,canny,laplaciann.py | sobel,canny,laplaciann.py | 7 months ago |

README License

# EDGE-DETECTION

## Aim:

To perform edge detection using Sobel, Laplacian, and Canny edge detectors.

## Software Required:

Anaconda - Python 3.7

## Algorithm:

### Step1:

Import all the necessary modules for the program.

### Step2:

Load a image using imread() from cv2 module.

### Step3:

Convert the image to grayscale

### Step4:

Using Sobel operator from cv2,detect the edges of the image.

### Step5:

Using Laplacian operator from cv2,detect the edges of the image and Using Canny operator from cv2,detect the edges of the image.

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### Code:

## Original:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

image = cv2.imread('spyd.jpg')
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title('Original Image')
plt.axis('off')
```

## SOBEL EDGE DETECTOR

```
sobel_x = cv2.Sobel(gray_image, cv2.CV_64F, 1, 0, ksize=5)
sobel_y = cv2.Sobel(gray_image, cv2.CV_64F, 0, 1, ksize=5)
sobel_combined = cv2.magnitude(sobel_x, sobel_y)
plt.imshow(sobel_combined, cmap='gray')
plt.title('Sobel Edge Detection')
plt.axis('off')
```

## LAPLACIAN EDGE DETECTOR

```
laplacian = cv2.Laplacian(gray_image, cv2.CV_64F)
plt.imshow(laplacian, cmap='gray')
plt.title('Laplacian Edge Detection')
plt.axis('off')
```

## CANNY EDGE DETECTOR

```
canny_edges = cv2.Canny(gray_image, 50, 150)
plt.imshow(canny_edges, cmap='gray')
plt.title('Canny Edge Detection')
plt.axis('off')
```

## Output:

### Original:

```
Out[1]: (-0.5, 187.5, 267.5, -0.5)
```

Original Image

SOBEL EDGE DETECTOR

```
Out[2]: (-0.5, 187.5, 267.5, -0.5)
```

Sobel Edge Detection

LAPLACIAN EDGE DETECTOR

```
Out[3]: (-0.5, 187.5, 267.5, -0.5)
```

Laplacian Edge Detection

## CANNY EDGE DETECTOR

```
Out[4]: (-0.5, 187.5, 267.5, -0.5)
```

## Result:

Thus the edges are detected using Sobel, Laplacian, and Canny edge detectors.

### Releases

No releases published

### Packages

No packages published

### Languages

- **Jupyter Notebook** 99.9%
- **Python** 0.1%